

Lecture 1: Introduction

HUM 4441: Engineering Ethics

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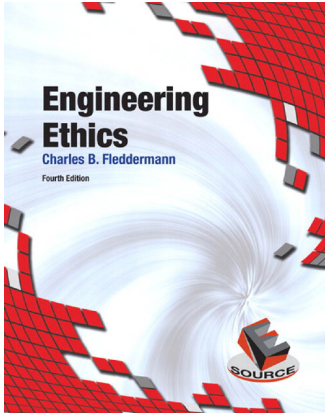
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Course Material



Primary Text

Charles B. Fleddermann

Engineering Ethics

- main source for chapter structure, concepts, and case studies
- use this book first when reviewing course material

Supplementary Text

Charles E. Harris Jr. et al.

Engineering Ethics: Concepts and Cases

- extra cases and alternative discussions
- useful when a topic needs a second angle

Evaluation Procedure

Coursework

Component	Marks
Assignment 1	15
Assignment 2	15
Presentation	30
Coursework total	60

Written Exams

Component	Raw marks
Mid Semester	120
Semester Final	120

Total Marks: 100

Raw exam scores are scaled into the course total.

The Ford Pinto Case



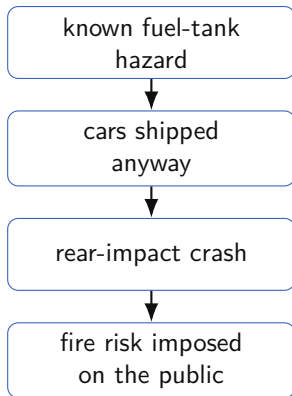
Ford Pinto subcompact. Rear-impact fuel-tank vulnerability made this car a standard ethics case.

In 1978, a Ford Pinto was hit from behind on an Indiana highway. The fuel tank ruptured, the car caught fire, and three teenage girls were killed.

The ethical force of the case is not the crash alone. The design vulnerability was already known. Once a company knows a danger and keeps shipping the product, safety, cost, speed, and responsibility are already in conflict.

Image: Wikimedia Commons; case discussed in Fleddermann, Ch. 1

Before Any Cost-Benefit Math, What Is Already Clear?



- The defect sits inside an ordinary product that users cannot inspect for themselves.
- The public faces the danger before it even understands that the danger exists.
- Once the hazard is known, every additional shipment is an active decision about who absorbs the risk.

If you knew about the fuel-tank risk, what would you insist on documenting before the next car shipped?

Why Does Engineering Ethics Matter?



Fragments recovered after the Challenger launch failure in January 1986.



Recovered Columbia debris examined during the 2003 investigation.

- Engineering decisions can kill, protect, or quietly expose people to risk.
- Failure is rarely only technical. Management, communication, and culture matter too.
- Public trust can collapse long after a design choice first looked routine.
- Ethics matters most when cost, schedule, and authority push against caution.

Core tension: the problem is rarely whether safety matters. The problem is what happens when safety collides with cost, authority, reputation, or speed.

What Are We Actually Studying?

Ethics

moral judgment about how people ought to act toward others

Engineering Ethics

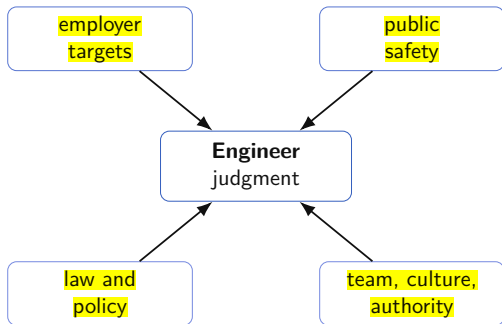
professional judgment inside technical work, institutions, and public risk

Main Focus

decisions, duties, uncertainty, failure, and responsibility

Engineering ethics starts when a technical decision changes what happens to other people and creates a duty to answer for it.

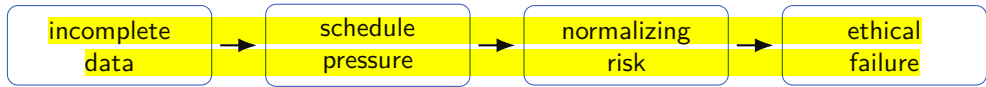
Moral Autonomy Under Pressure



Moral autonomy means holding onto professional judgment even when the surrounding system rewards silence, speed, or obedience.

- spotting obvious right and wrong is not the hard part
- the hard part is keeping judgment intact under pressure
- professional ethics begins where passive obedience stops

Why Good Intentions Are Not Enough



Many failures begin with competent people making decisions that feel practical at the time. Good intentions do not cancel weak judgment, weak process, or weak resistance to pressure.

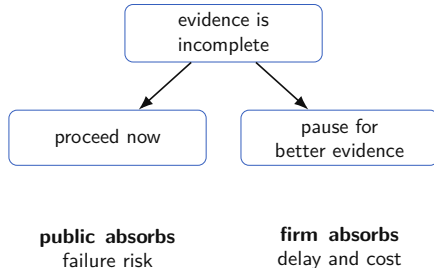
Engineering Means Managing the Unknown

- How well will this work outside ideal conditions?
- Who carries the risk if our estimate is wrong?
- What changes if the system is scaled, copied, or widely adopted?
- What happens if we continue without better evidence?

Good practice

- test seriously
- document uncertainty
- anticipate failure
- resist fake certainty

If the Evidence Is Incomplete, Who Carries the Burden?



- **Uncertainty does not remove responsibility.**
- Someone still absorbs the consequences of acting too early.
- Ethical judgment asks who is being asked to carry that risk, and why.

When the data are weak, should the burden of proof fall on proceeding or on pausing?

Personal Ethics and Professional Ethics

Dimension	Personal ethics	Professional ethics
Focus	everyday conduct with other people	duties inside a formal role with public consequences
Setting	private and social life	employer, regulator, client, and public
Pressure	mostly personal	institutional, legal, economic, and technical
Stakes	limited local harm	large-scale harm, system failure, loss of trust

Ethics and the Law

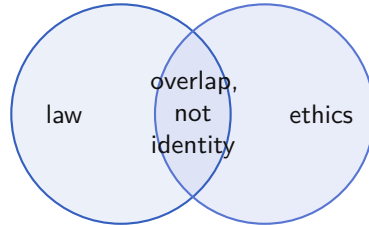
Legal but still unethical

A product can meet the letter of the law and still expose people to avoidable danger.

Illegal but not obviously unethical

Law can lag behind technical evidence, especially when regulation moves slowly.

Takeaway: law sets a floor. Ethics matters most when the law is silent, late, narrow, or technically shallow.



Ethical Problems Behave More Like Design Problems

Why the comparison matters

- both involve constraints, trade-offs, and incomplete information
- both require judgment rather than a single mechanical answer
- both still allow stronger and weaker responses

What does not follow

- ambiguity does not mean anything goes
- disagreement does not make all positions equally defensible
- uncertainty increases the need for judgment; it does not remove it

The Space Shuttle Challenger



Fragments recovered after the Challenger explosion in January 1986.

Engineers had already seen warning signs in the solid rocket booster joints. On the night before launch, they argued that the shuttle should not fly in the expected cold weather. The launch went ahead anyway. Cold conditions worsened the O-ring failure, hot gases escaped, and the shuttle was destroyed.

This case matters because the warnings were not absent. They were argued over, downgraded, and overridden.

NASA aftermath image; Challenger discussed in Fleddermann, Ch. 1

What Was Known Before Launch?

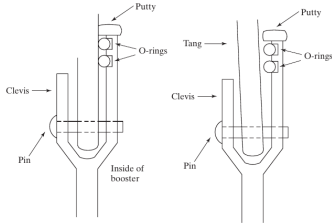


Figure 1.1
[a] A schematic drawing of a tang and clevis joint like the one on the Challenger solid rocket boosters.
[b] The same joint as in Figure 1.1a, but with the effects of joint rotation exaggerated. Note that the O-rings no longer seal the joint.

- O-ring erosion was already documented.
- Cold weather increased concern about sealing performance.
- The strongest warning was not lack of data. It was the choice to **treat incomplete data as permission to proceed.**

If the evidence is worrying but not mathematically complete, is that enough to stop a launch?

Figure 1.1: the booster-joint schematic shows how joint rotation could prevent the O-rings from sealing.

Book figure from Fleddermann, Ch. 1

Columbia and the Danger of False Confidence



Investigators examine recovered Columbia debris after the 2003 disaster.

Columbia adds a different warning. Foam strikes had happened before. Because earlier missions survived, the organization started treating the hazard as normal.

- repeated survival can create false confidence
- visible danger can slowly become routine
- one tragedy does not automatically fix culture

If Safety Advice Is Ignored, What Comes Next?

Once a serious warning is dismissed, the problem changes. The question is no longer only whether the design is safe. The question becomes how far professional responsibility extends after disagreement.

- document the concern clearly
- escalate through the available chain
- refuse direct participation when necessary
- consider outside disclosure only when the risk is serious and internal channels fail

The hard part is not noticing the problem.

The hard part is deciding what you still owe the public after you have been overruled.

Lecture 1 Summary

- Engineering ethics is about professional judgment under pressure, uncertainty, and real consequences.
- The hard cases are not clean. They involve trade-offs, weak evidence, hierarchy, and institutional momentum.
- Law matters, but it does not settle the whole ethical problem.
- Professional ethics is broader than private morality because the stakes and structures are different.
- The Ford Pinto, Challenger, and Columbia cases all show the same pattern: warning signs exist, but organizations still find ways to move ahead.

References

- Charles B. Fleddermann, *Engineering Ethics*, 4th ed., Chapter 1.
- Charles E. Harris Jr. et al., *Engineering Ethics: Concepts and Cases*.